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Experiments and analysis on LBE steady natural circulation in a rectangular shape loop

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ABSTRACT

For the purpose of investigating the thermal-hydraulic characteristics of lead-bismuth eutectic (LBE) natural circulation (NC), a LBE NC loop (NC-loop), experimental platform of the KYLIN-II facility was operated by the Institute of Nuclear Energy Safety Technology (INEST) in 2011. Experiments to characterize the thermal-hydraulic performance of HLM coolants undergoing NC were carried out at steady state to study the following issues: 1) the initiation of NC at different power levels; 2) the effects of input power and secondary loop operational characteristics on NC; 3) the capacity limits of NC and 4) a comparison of experimental data with theoretical predictions. The maximum NC capacity for LBE with a flow rate of 1.2 kg/s was obtained for a 24 kW heat pin corresponding to heating power input of 200 W/cm, which is higher than the linear power rating of the proposed China LEad Alloy Reactor-I (CLEAR-I). The experimental results at high heat pin input power were found to be in good agreement with the theoretical predictions.

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1. Introduction

Heavy liquid metals (HLM) such as pure lead and its alloys are chosen as the main candidate coolants for future advanced nuclear systems due to their good nuclear and thermalphysical properties. Especially, lead-bismuth eutectic (LBE) has been chosen as the material for spallation target and sub-critical core coolant for accelerator driven system (ADS) (Rubbia, 1996). Due to their high thermal expansion coefficients, HLM-cooled nuclear systems have a significant NC capability (Cheng et al., 2004) for improving the passive safety of a reactor. Some advanced HLM-cooled reactors designs even apply NC as normal operation mode, e.g. the SSTAR (Smith, 2012). Moreover, NC could be greatly enhanced by injecting an inert gas (e.g. Argon gas) at certain pressure condition into the hot leg of the HLM coolant circuit as proposed by XADS (Cinotti and Gherardi, 2002), for which is an experimental accelerator driven system.

An ADS project was launched by Chinese Academy of Sciences in 2011. Under this project, a ~10 MW LBE cooled reactor named CLEAR-I was proposed by FDS team. Meanwhile, KYLIN-II, a large scale multi-functional LBE experimental facility for materials corrosion, thermal-hydraulics and safety study related to CLEAR-I, was built in 2013 by FDS team and then operated to verify the reactor design (Wu et al., 2010). FDS team also has long been devoting into fusion reactor design (Wu and FDS Team, 2006, 2008; Wu et al., 2006, 2011), R&D on the blanket (Wu and FDS Team, 2006, 2007), R&D on fusion materials and technologies (Huang et al., 2004, 2009), neutronics (Wu et al., 1999; Wu and FDS Team, 2009), and so on.

Although LBE-cooled reactors based on NC has many attractive advantages compared with traditional reactors utilizing mechanical pumps, some great challenges and uncertainties should be studied due to the lack of operation experiences. However, very few experiments have been carried out so far for LBE NC. In general, almost all the experiments results got until now show that LBE NC can be created quickly and easily in a short time, and then reach steady state in ~30 min from initial zero power. In addition, Ma et al.'s experiment results (2007) showed that an unstable NC was obtained at specific conditions. Tarantino and Bernardi (2010) performed LBE NC tests with gas-injection on the NACIE loop and CIECE pool test facility to verify the code post-test in ENEA. Moreover, the stable LBE NC had been obtained from experiments by Takahashi (2005), Choa and Batta (2011) and Borgohain and Jaiswal (2011) performed at different conditions.

Some experiments mentioned above were carried out for HLM NC study. However, there were few studies for NC at

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continuously increased or decreased input power, and the maximum input linear power rate was 130 W/cm. In our experiments, the experiment for NC creating process and the effect of the parameters are carried out in NC-loop. The NC thermal-hydraulics characteristics are also studied with various input powers of heat pin among 5 kW and 24 kW. Moreover, the maximum NC capacity at 200 W/cm linear power rating level had been obtained based on the loop. In addition, the experimental data can be used to verify and qualify numerical codes, such as the RELAP5 code.

2. Natural circulation loop (NC-loop)

NC-Loop is a sub-loop of KYLIN-II with a rectangular shape (as seen in Fig. 1). The loop mainly consists of two vertical pipes serving as riser and down-comer respectively, two horizontal pipes collecting the riser and down-comer, an expansion tank at the top of the hot leg, a heat source (HS) at the lower part of the riser, a heat exchanger (HX) at the upper part of the down-comer and a cooling secondary loop installed at the other side of the HX. Glycerinum oil is selected as coolant of the secondary loop, because it's allowable temperature can reach ~300 °C. The loop is also designed to reserve a joint for gas injection at the downstream of the HS. Table 1 shows the main parameters of the loop. The loop is foreseen to have three operation modes:

a) natural circulation;
b) gas-lift driven circulation;
c) gas-injection enhanced circulation.

Table 1
Design parameters of the NC-loop.

Item	Parameter
Total height (m)	5.0
Length of HS (m)	1.8
Length of HX (m)	1.5
Length of gas-lift riser (m)	3.0
Pipe inner diameter (cm)	2.6
Operating temperature (°C)	150–500
Operating pressure (atm.)	1–1.5
Input power (kW)	0–24

All pipes and components in the loop, which contact directly with the high temperature liquid LBE, are made of 316 L, while the secondary pipes in contact with oil are made of 304 stainless steel. The loop is filled with argon gas at ~0.15 MPa before filling LBE from store tank. The maximum LBE design temperature is 500 °C, while the normal operation temperature are maintained at ~450 °C. A 24 kW heat pin is used to be heat source with 200 W/cm linear power level over core fuel pin in CLEAR-I. To avoid overheat for the coil of heat pin, a cooling fan is placed at bottom of the NC-loop. LBE is currently employed as working fluid of the primary loop. A tube in tube HX with secondary oil flowing in out-pipe is set at the upper part of the downcomer with the maximum power of 30 kW.

In order to measure the NC-loop temperature accurately, four immersion thermocouples were installed at the both ends of the HX and HS. The flow rate of LBE can be obtained by the power balance calculation method in the experiments. Two level meters are set in the expansion tank at different levels. And the higher one indicates alarm level, while the lower one indicates operation level.

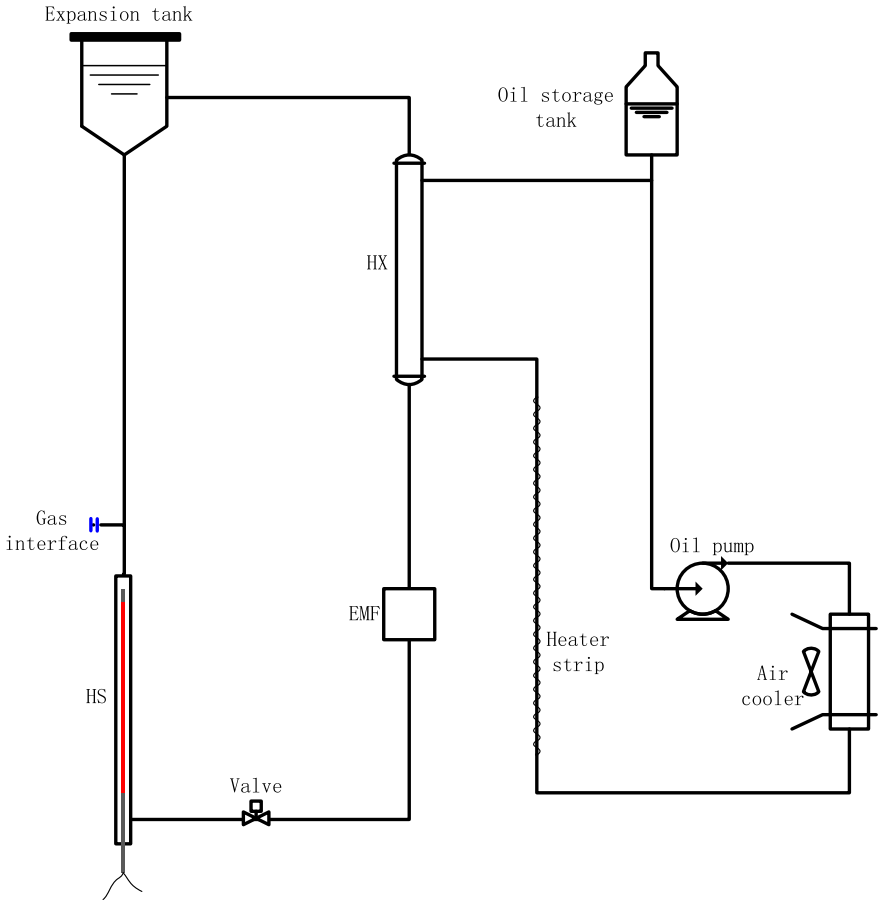


Fig. 1. Schematic of NC-loop of KYLIN-II.

The secondary oil flow rate controlled by a transducer of oil pump is measured with a vortex flow meter installed in front of the input the HX.

3. Results and analysis

3.1. Natural circulation start-up

The NC is created through the heating and cooling in the primary loop, which is a dynamic process. In a prototypical reactor, which using NC as normal operation mode, e.g. the SSTAR, it can occur during the reactor start-up. Moreover, the NC can be created also in a forced circulation reactor, when the driving force is lost (e.g. malfunction of the pump). So it is important to know LBE thermal-hydraulics characteristics during the NC start-up.

Thus, in the first case, Fig. 2 shows the process of NC created from zero power conditions to 20 kW. The initial conditions of the loop are shown in the following. 1) The temperature of LBE is not uniform. The average temperature of the hot leg is higher than that of the cold leg as shown in Fig. 2(a). 2) The secondary loop started for operation when the temperature of oil reaches to 180 °C and the flow rate to 1.02 kg/s 3) The auxiliary heating system is running, while the power of heat pin is 0. Because of the auxiliary heating system and secondary loop operation, an NC has been created slightly between hot and cold legs.

The case begins by suddenly increasing of the input power as shown in Fig. 2(d). Fig. 2(a) shows the temperature variation during the creating process. The temperature measured with the four immersion thermocouples increase rapidly and the temperature difference between hot and cold legs in a short time as seen in Fig. 2(b). Moreover, there is a peak value appeared at ~80 s after the power input.

As seen in Fig. 2(c), strong LBE NC is observed with the increase in flow rate. The circulation flow rate increases quickly from zero to peak during ~3 min. The main reason is that LBE flows in the loop

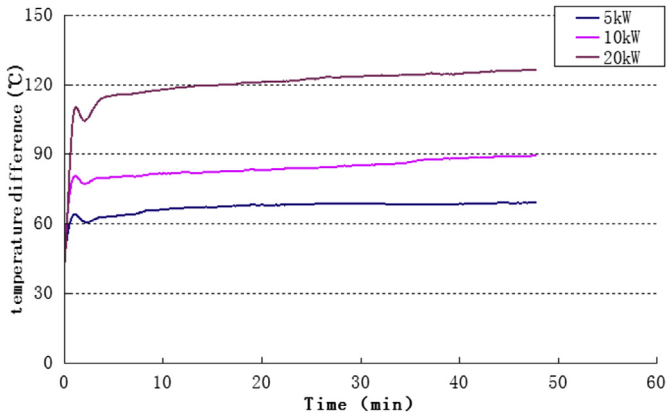


Fig. 3. Variation trend of temperature difference between hot and cold leg in nature circulation creating at different power levels.

heated by the pin first. And the LBE from the pin moves upwards to the HX. So the delay time for the measurement between the hot and cold legs leads to the peak value. Later, all these parameters become stable, and steady state conditions are created corresponding the steady state natural circulation is obtained in about half an hour. Fig. 3 shows the variation processes of temperature difference between hot and cold legs at different power levels. Obviously, the temperature difference during NC created processes are similar at different power levels.

3.2. Effect of parameter variation

Usually, the NC capacity depends on temperature difference between hot and cold legs. And the temperature difference is strongly affected by the power of heat source and flow rate of the secondary loop. In the following sections, the effects of the two factors have been investigated.

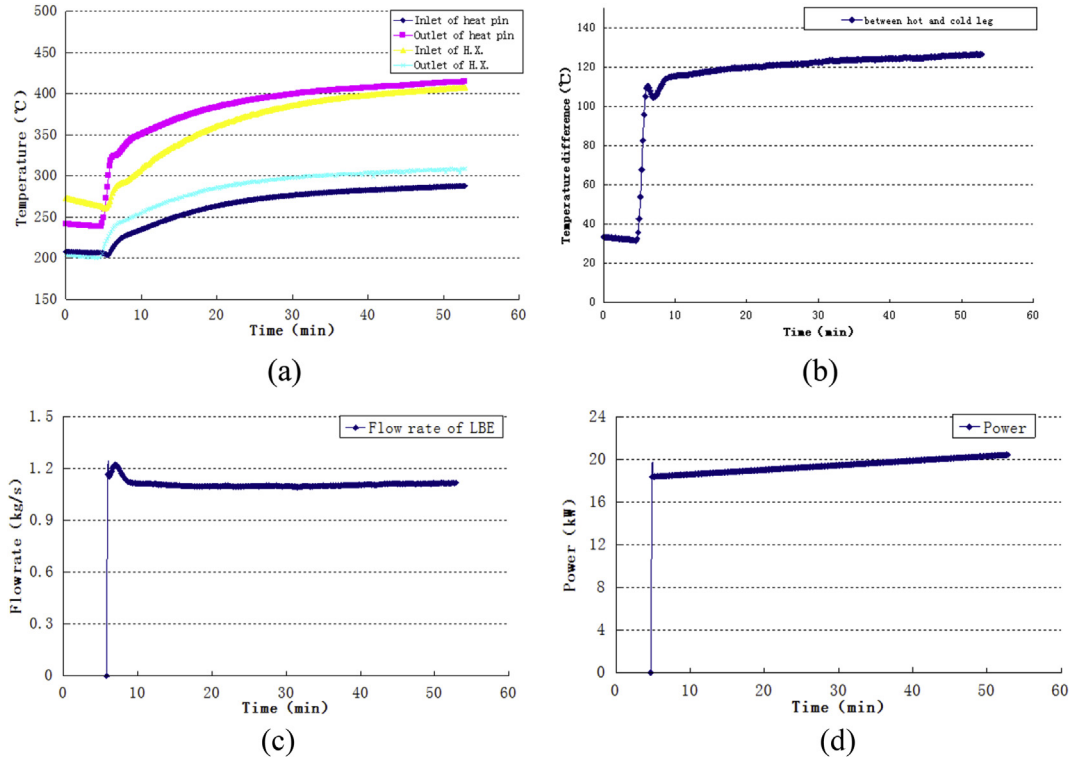


Fig. 2. The process of NC created from zero power to 20 kW conditions.

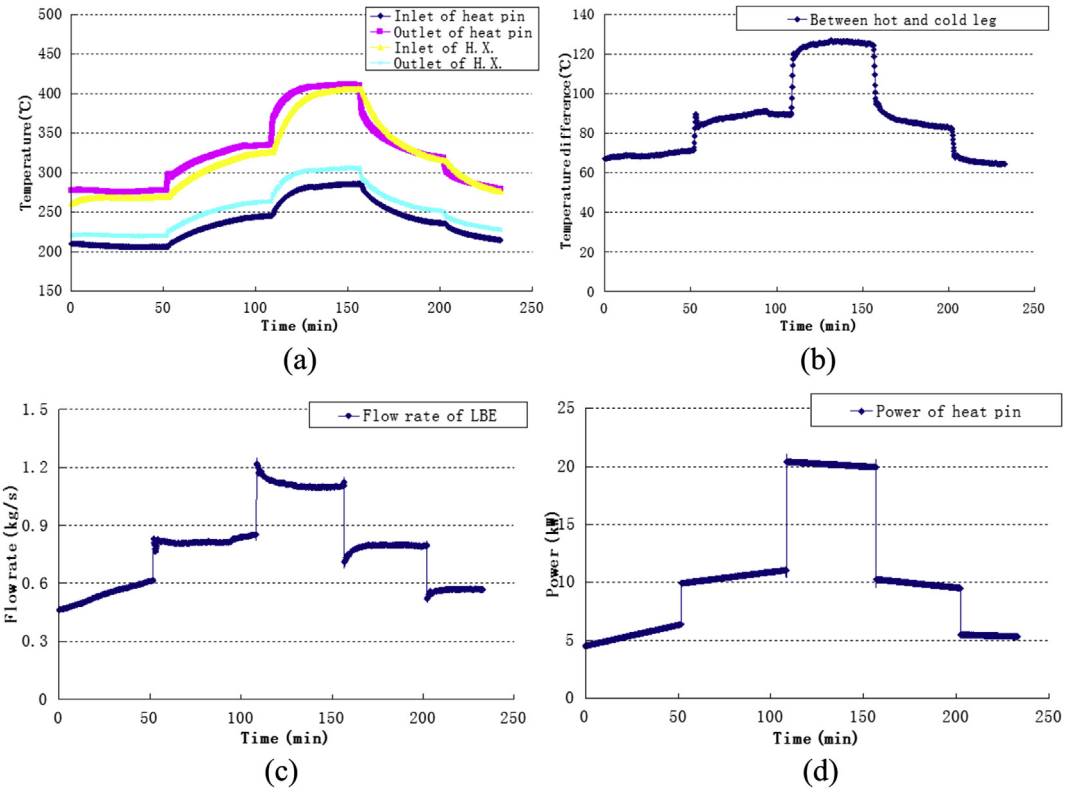


Fig. 4. Parameters variation with the change of input power step by step.

3.2.1. *Input power*

In the second case, Fig. 4 shows the effect of the power on the NC for three power level, while keeping the same secondary flow rate i.e. 1.02 kg/s. The initial conditions of the loop are stable state NC created at ~5 kW. The temperature of the four immersion thermocouples increases with the increased input power as seen in Fig. 4(a), while Fig. 4(b) shows the corresponding temperature difference between hot and cold legs. A new steady NC is created

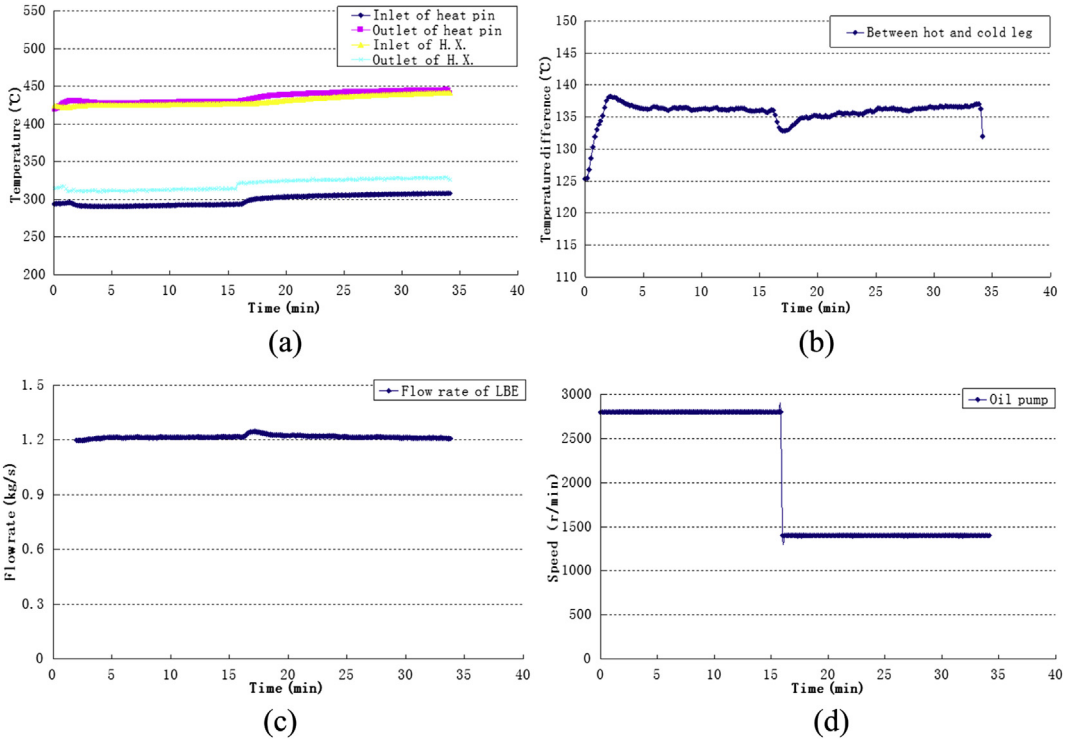


Fig. 5. Parameters variation with the change of secondary pump speed.

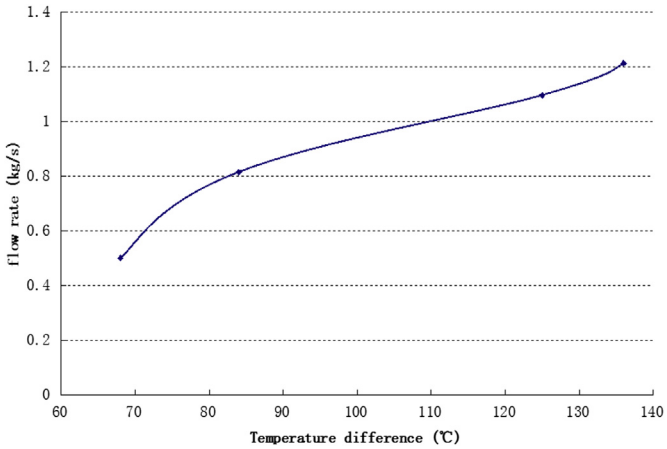


Fig. 6. The temperature difference and corresponding flow rate.

quickly with stable temperature difference and flow rate as shown in Fig. 4(c). The temperature difference changes from 68 °C to 89 °C and 125 °C at three input power levels, while the corresponding flow rate is 0.5 kg/s, 0.8 kg/s and 1.1 kg/s respectively. Moreover, the temperature difference during the power increasing is slightly lower than that during power decreasing. In the whole experiment process, no instability was observed at any conditions.

3.2.2. Flow rate of secondary loop

In the third case, the flow rate of oil is employed to study the effect of the NC. In the NC process, the secondary flow rate affects the energy transport for the primary. In the case, the flow rate of oil is decreased with reduced half of speed of the pump, while keeping the other parameters constant. Fig. 5 shows the effect of the flow rate of oil on the NC. The initial conditions of the loop are that a stable state NC has been created at ~24 kW. The oil flow rate is decreased suddenly from 1.0 to 0.5 kg/s with reduced half of speed of the pump as seen in Fig. 5(d). The temperature of the four immersion thermocouples affects directly and increases quickly, when the flow rate of oil decreases. As shown in Fig. 4(b), the decreased temperature difference of cold and hot legs increases about 3 °C after the flow rate decreased in about 2 min, and then backs to the previous level quickly. Gradually, the temperature difference almost returns to their initial values. The reason for apparent increase of the temperature of oil in secondary loop can be analyzed about the conservation of energy between LBE and oil. Due to the almost no change for the heat transfer coefficient of the oil, the effect of NC on the flow rate of the oil is not obvious.

3.3. Capability of natural circulation

It is important to obtain greater capacity on NC in a reactor design and operation. In the reactors, the greater capacity means more security features. In order to obtain the capacity of NC in the loop from the experimental results, the temperature difference and its corresponding flow rate are shown in Fig. 6. At maximum temperature difference and the valve opening, the circulation flow rate is 1.21 kg/s under input power 24 kW, while the corresponding secondary oil flow rate is ~1.0 kg/s. This power level corresponding the heat input of ~200 W/cm of the heat pin is higher than the linear power rating of CLEAR-I as an experimental ADS design. This also implies that the NC can be widely used in more types reactors design by using LBE or other HLM coolants due to its significant NC capability.

3.3.1. Comparison with formulas calculation

Formulas (Ran et al., 2013) were employed to know the natural circulation capacity for the NC-loop design before the experiment performed. Comparisons between experimental data and formulas calculation are performed to validation of the formulas feasibility in the HLM system design. Fig. 7 shows the formulas calculation about the flow rate (a) and temperature difference (b) compared with the experimental data at different input powers. Obviously, the NC flow rate from experiment is underestimated compared to the correlation, while the temperature difference is overestimated. As shown in Fig. 7, the flow rate as well as the temperature difference is consistent very well at high input power between experimental results and the formulas calculation. But the flow rate of the calculation is lower than that of experimental results at 5 kW power, while the temperature difference is higher than 26%. The main reason is that the heat dissipation of cold leg is higher than that of the hot leg because of a cooling fan placed at bottom of the NC-loop to avoid overheating for the coil of heat pin. So, the heat dissipation effect during NC cannot be ignored at low input power.

4. Conclusion

The tests on the LBE natural circulation have been carried out in NC-loop of KYLIN-II. The main results are summarized as following:

- Due to LBE high thermal expansion property, natural circulation is quickly created in the HLM system. The temperature difference between hot and cold legs as well as the flow rate increase quickly and reach to the maximum value in ~2 min. And in the range of half an hour, a steady circulation can been performed.

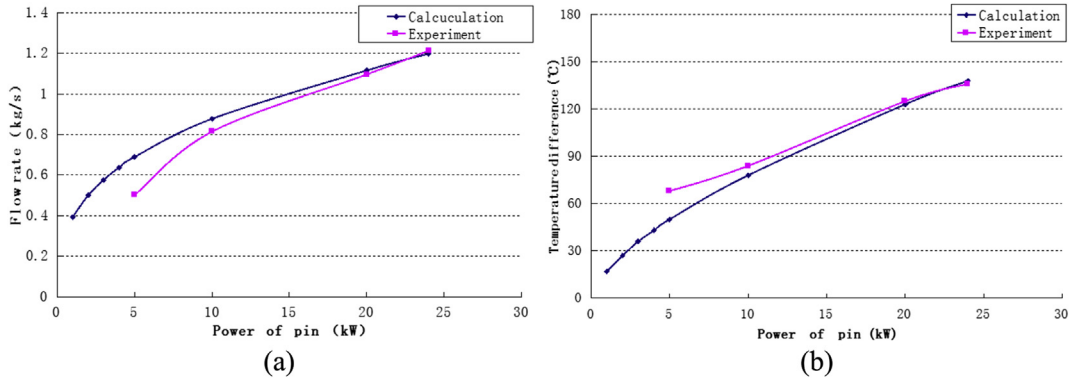


Fig. 7. Comparisons between the formulas' calculation with the experimental results at different input power.

- After a steady circulation created, it have been performed to study parameters effects on the NC, for what steep variations one of the parameters (input power, secondary flow rate) with keeping other parameters constant during the experiments. The NC is affected strongly by the input power and weakly by the secondary flow rate.
- The max. circulation capacity in the experiments, when the circulation flow rate is 1.21 kg/s, can remove 200 W/cm linear power rating from the heat pin by NC-loop. This implies that the NC can be widely used in more types reactors design by using LBE or other HLM coolants due to its significant NC capability.
- The flow rate as well as the temperature difference is consistent very well at high input power between experimental results and the formulas calculation. But the flow rate of the calculation is lower than that of experimental results at 5 kW power, while the temperature difference is higher than 26%.

Further analysis and experiments will be carried out, such as analysis of RELAP5, transient experiments, and so on.

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